

Project: Running Nmap Port Scans

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First, set up your VM in Kali Linux. Update the machine and configure it to the latest upgrade using the command: "`sudo apt upgrade`". Then get the latest update package using the command: "`sudo apt-get update`"

Then, get your IP address that you want to begin scanning (i.e, 192.168.x.x)

```
(mallikah@kali)-[~]
└─$ sudo apt update
[sudo] password for mallikah:
Get:1 http://mirror.ourhost.az/kali kali-rolling InRelease [41.5 kB]
Get:2 http://mirror.ourhost.az/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://mirror.ourhost.az/kali kali-rolling/main amd64 Contents (deb) [50.8 MB]
Get:4 http://mirror.ourhost.az/kali kali-rolling/contrib amd64 Packages [119 kB]
Get:5 http://mirror.ourhost.az/kali kali-rolling/non-free amd64 Packages [201 kB]
Get:6 http://mirror.ourhost.az/kali kali-rolling/non-free-firmware amd64 Packages [10.6 kB]
Fetched 72.1 MB in 4min 3s (296 kB/s)
38 packages can be upgraded. Run 'apt list --upgradable' to see them.

(mallikah@kali)-[~]
└─$ sudo apt-get upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following packages were automatically installed and are no longer required:
  fonts-inter-variable gnome-accessibility-themes gnome-themes-extra gnome-themes-extra-data icu-devtools libflac12t64 libgail-common
  libgail18t64 libgeos3.13.0 libglapi-mesa libgtk2.0-0t64 libgtk2.0-bin libgtk2.0-common libicu-dev liblbfgsb0 libpoppler145
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 python3-setproctitle python3.12-tk ruby-zeitwerk
Use 'sudo apt autoremove' to remove them.
The following packages have been kept back:
  bulk-extractor gir1.2-javascriptcoregtk-4.1 gir1.2-webkit2-4.1 gstreamer1.0-plugins-bad kali-desktop-gnome libges-1.0-0
  libgstreamer-plugins-bad1.0-0 libjavascriptcoregtk-4.1-0 libre2-11 libwebkit2gtk-4.1-0 python3-fonttools
The following packages will be upgraded:
  apt apt-utils gvimd gvimd-common intel-media-va-driver libapt-pkg7.0 libcdio19t64 libegl-mesa0 libgbm1 libgl1-mesa-dri libglx-mesa0
  libspeechd-module0 libspeechd2 libsvn1 libxatracker2 mesa-libgallium mesa-va-drivers mesa-udpau-drivers mesa-vulkan-drivers
  metasploit-framework python3-flask-sqlalchemy python3-paho-mqtt python3-speechd speech-dispatcher speech-dispatcher-audio-plugins
  speech-dispatcher-espeak-ng subversion
```

```
(mallikah@kali)-[~]
└─$ ip address
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2e:1a:8f brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84318sec preferred_lft 84318sec
```

1. Simple Host Discovery

Command: `nmap -sn [Target IP or range]`

Flag Explanation: The `-sn` flag initiates a simple host discovery (ping scan) to check which hosts are up without scanning ports.

```
(mallikah@Kali)-[~]  
$ nmap -sn 10.0.2.15  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 02:44 EDT  
Nmap scan report for 10.0.2.15  
Host is up.  
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

Why It's Used: This scan is primarily used to identify live hosts on a network without triggering a more intrusive port scan. It's often the first step to establish a baseline of active devices in an environment.

Information to Look For:

- List of live hosts.
- Unexpected devices on the network (indicating unauthorized access or rogue devices).

SOC Relevance: Understanding which devices are live helps to maintain an accurate asset inventory and identify suspicious or unauthorized devices that shouldn't be on the network. This step is crucial for threat hunting and asset discovery in a SOC environment.

2. Basic Port Scan

Command: `nmap -p 80,443 [target IP]`

Flag Explanation: The `-p` flag specifies which ports to scan (e.g., ports 80 and 443 for HTTP and HTTPS).

```
(mallikah@Kali)-[~]  
$ nmap -p 80,443 10.0.2.15  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 02:52 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.00042s latency).  
  
PORT      STATE SERVICE  
80/tcp    closed http  
443/tcp    closed https  
  
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

Why It's Used: This scan targets specific ports to identify if common services, like web servers, are accessible on a host. This can reveal what services are running and whether these ports are unexpectedly open.

Information to Look For:

- Open ports that indicate running services.
- Services that shouldn't be accessible externally (e.g., database or management interfaces).

SOC Relevance: In a SOC role, you need to verify that only authorized services are accessible on a network. If critical ports like SSH (22) or RDP (3389) are exposed when they shouldn't be, it might indicate a misconfiguration or a security gap.

3. Service Version Detection

Command: `nmap -sV [target IP]`

Flag Explanation: The `-sV` flag attempts to detect the versions of services running on open ports.

```
(mallikah@Kali)-[~]
$ nmap -sV 10.0.2.15
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 02:54 EDT
Nmap scan report for 10.0.2.15
Host is up (0.0000010s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.43 seconds
```

Why It's Used: Service version detection helps identify the exact version of software running on a target. This is essential for vulnerability assessment, as specific versions may have known vulnerabilities.

Information to Look For:

- Service names and version numbers.
- Deprecated or vulnerable software versions.

SOC Relevance: Accurate version information is vital in a SOC role to correlate services with known vulnerabilities. For example, detecting an outdated version of Apache or a vulnerable FTP service would require immediate attention and remediation to prevent potential exploits

4. Full TCP Scan

Command: `nmap -sT [target IP]`

Flag Explanation: The `-sT` flag initiates a full TCP connection scan, providing accurate results but is noisier.

```
(mallikah@Kali)-[~]  
$ nmap -sT 10.0.2.15  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 02:58 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.00049s latency).  
All 1000 scanned ports on 10.0.2.15 are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

Why It's Used: A full TCP scan (also known as a connect scan) establishes a complete TCP handshake with each port, making it more reliable in detecting open services compared to a SYN scan.

Information to Look For:

- Comprehensive list of open ports and running services.
- High number of unexpected open ports.

SOC Relevance: In a SOC environment, a full TCP scan can help map the complete attack surface of a device. This information is useful for conducting detailed security assessments and identifying potential attack vectors.

5. Aggressive Scan

Command: `nmap -A [target IP]`

Flag Explanation: The `-A` flag enables aggressive scanning, combining multiple advanced scanning techniques into one command.

```
(mallikah@Kali)-[~]  
$ nmap -A 10.0.2.15  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 03:04 EDT  
Nmap scan report for 10.0.2.15  
Host is up (0.00018s latency).  
All 1000 scanned ports on 10.0.2.15 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.02 seconds
```

Techniques Included:

- **Operating System Detection (-O):** Attempts to identify the OS of the target, providing insight into the system type and potential vulnerabilities.
- **Service Version Detection (-sV):** Determines the versions of services running on open ports, useful for vulnerability identification.
- **Script Scanning (-sC):** Runs a set of default Nmap Scripting Engine (NSE) scripts for vulnerability detection, like checking for misconfigurations or weak authentication.

- **Traceroute:** Maps the network route to the target, useful for identifying intermediary devices and understanding the network topology.

Why It's Used: This scan is ideal for gathering a comprehensive view of a target in one command, especially during a deep assessment or when facing time constraints.

Information to Look For:

- Detailed service and version information.
- Operating system and network layout.
- Script output that highlights potential vulnerabilities.

SOC Relevance: Aggressive scans can be useful in a SOC for thorough security assessments or to identify detailed information during incident response. However, they should be used cautiously in a production environment as they generate a lot of noise and could trigger security alerts.

Summary

Each of these scans serves a distinct purpose for a SOC analyst:

1. **Host Discovery** is used to identify the presence of assets and potential threats.
2. **Basic Port Scans** reveal service exposure that could be exploited.
3. **Service Version Detection** maps the services to specific vulnerabilities.
4. **Full TCP Scans** ensure a complete understanding of the network's attack surface.
5. **Aggressive Scans** gather a holistic view of the system's state, useful for detailed assessments or investigations.